Cytoprotective Action of Folic Acid Supplementation in Experimentally-Induced Gastric Ulceration in Rats

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ABSTRACT

The effect of folic acid (2 mg/kg oral) on experimentally induced gastric ulcers was investigated by evaluating gastric ulcer indices, acid secretion and gastric biochemical levels of protein, markers of oxidative stress and antioxidant, mucin and nitric-oxide in adult male Wistar rats. Twenty rats were divided into 4 groups as indomethacin ulcer control, folic acid -treated + indomethacin ulcer, pyloric ligation control and folic acid -treated + pyloric ligation ulcer. Rats treated with the drug of study were pre-treated for 6 days and one hour before ulcer induction. The acute gastric ulcer models used were indomethacin induced and pylorus ligation and were performed following standard laboratory methods. Four hours after ulcer induction, laparectomy was performed and the stomach harvested for macroscopic and acid content studies and biochemical analysis. The data was analyzed using statistical tools from SPSS (version 21) and "student t test" was performed at $p<0.05$ level of significance. Folic acid treatment resulted in decreased in ulcer index with a protective ratio of 55.56%, non-significant increase in gastric acidity, increase in gastric protein (1.07±0.06 vs 1.28±0.13; $p>0.05$), nitric oxide (120.33±6.67 vs 151.17±28.16; $p<0.05$), mucin content (36.20±1.99 vs 36.89±4.29; $p>0.05$) and GSH levels (2.38±0.40 vs 2.51±0.18; $p>0.05$) and decrease in MDA (1.16±0.001 vs 0.59±0.003; $p>0.05$) compare to control in the indomethacin induced ulcer model. Folic acid treatment in the pyloric ligation ulcer model resulted in a significant decrease in ulcer index with a protective ratio of 60.00%, non-significant increase in gastric acidity, mucin content (38.58±3.03 vs 39.55±0.84; $p>0.05$), but decrease in gastric MDA (1.19±0.01 vs 0.89±0.001; $p>0.05$), GSH (2.49±0.29 vs 2.36±0.23; $p>0.05$), nitric oxide (145.89±14.86 vs 106.86±7.64) and protein (1.31±0.01 vs 1.21±0.01) compared to control. The antiulcer activity of folic acid may be attributed to its stimulatory effect on gastric mucin concentration and inhibitory effect on oxidative stress.

Keywords: Gastric ulcer, Folic acid, Indomethacin, Pyloric ligation.

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1. INTRODUCTION

Gastric ulcer is a heterogeneous disorders manifesting as a break in the gastrointestinal mucosa lining[1]. It is the most predominant of gastrointestinal diseases with various epidemiologic data showing its steady incidence rate increase in different parts of the world[2-5]. It is also one of the major health hazard in terms of morbidity and mortality[6] affecting a considerable number of people worldwide[7]. Though its prevalence is high in Africa and Asia[8], the disorder is common throughout the World and in the past one or two decades there has been a phenomenal increase in the knowledge of its treatment[9]. Efforts in the search of gastro-protective agents also remain inexhaustible.

Many studies have documented the natural cause of gastric ulcer to be due to imbalance between the aggressive damaging factors like acid, pepsin, lipid peroxidation, refluxed bile, abnormal motility and the defense mechanisms, which protect the mucosa[10-12]. The exogenous aggressive factors like Helicobacter pylori, indiscriminate use of non-steroidal anti-inflammatory drugs (NSAIDs), alcohol abuse and smoking have also been identified as contributors[13]. The mucosal defense factors include mucus, antioxidants, surface-active phospholipids and

bicarbonates, the epithelium, the microcirculation, sensory afferent neurons beneath the epithelium, the mucosal immune system and regenerative ability of the mucosa[14].

Previous studies have increasingly shown that disturbances in gastric mucosal blood perfusion might cause ulcers and erosions[15] as lesions from ulcer and or gastric cancer can cause bleeding that will eventually lead to iron deficiency anemia (IDA). Multiple studies have implicated an association between *H pylori* infection and IDA[16-18]. The theories have been that *H pylori* infection can lead to IDA, including impairing iron absorption, competing with the host for uptake of iron, or elevating the pH and reducing vitamin C concentration[19-20]. Iron deficiency is a common nutritional disorder (estimated as about 500 million people worldwide)[21]. Since IDA is associated with high incidence of *H. pylori*[22], is it possible that diminishing IDA through folic acid supplementation may mitigate the aggressive factors of gastric ulcer formation?

Folate is a water-soluble B vitamin occurring naturally in large quantity in dark green leafy vegetables and citrus fruits[23]. Folate is generally regarded as safe and is assumed as an ideal functional food component for disease prevention[24]. Folic acid is the completely oxidized monoglutamyl form of folate that is used commercially in supplements and in fortified foods[25]. Its deficiency appears to play an important pathogenic role in anemia, atherosclerosis, adverse pregnancy outcomes, neuropsychiatric disorders, and cancer[26]. Its essential role in preventing neural tube defect has caused the inclusion of folic acid fortification of many foods and ready-to-eat breakfast cereals, and fruit juices[27-28]. Folate is known to plays a key role in DNA synthesis and methylation, and maintenance of DNA integrity, stability and repair[29]. Folate modulates a number of disorders such as cardiovascular diseases[30], neural tube and congenital defects[31], subfertility[32] and several malignancies like cancer of the colorectum, lungs, pancreas, esophagus, stomach, cervix, breast[33] and neuroblastoma and leukemia[26]. The mechanism by which folate modulates these disorders is related to its anti-apoptotic and anti-oxidative properties[26]. This study, investigates the protective effect of folic acid on gastric mucosal damage induced by indomethacin and pyloric ligation in rat model.

2. MATERIALS AND METHODS

2.1. Experimental animals: The study involves 20 adult Wistar rats (weighing between 120g and 150 g). They were obtained from the Animal Holding Facility of the College of Medicine, Ambrose Alli University, Ekpoma. They were then transferred to the experimental site in the Department of Physiology where they were maintained in plastic cages (with a wire net top) for 2 weeks of acclimatization under 12/12 hour light/dark conditions. They were allowed to feed *ad libitum* feeding on rat chow (Livestock Feed PLC, Ikeja, Lagos, Nigeria) and clean water and maintained according to procedures and Guidelines for Care and Use of Laboratory Animals in Biomedical Research[34].

2.2. Animal grouping: The rats were divided to 4 groups (n = 5) and treated as follow;

Group A1: indomethacin ulcer control group, made up of rats without pre-treatment but submitted for indomethacin-induced gastric ulcer

Group A2: indomethacin ulcer test group, made up of rats pre-treated with folic acid before submitted for indomethacin-induced gastric ulcer

Group B1: pyloric ligation ulcer control group, made up of rats without pre-treatment but submitted for pyloric ligation-induced gastric ulcer

Group B2: pyloric ligation ulcer test group, made up of rats pre-treated with folic acid before submitted for pyloric ligation gastric ulcer.

Pre-treatment of the A2 and B2 groups with 2 mg/kg folic acid was orally[35] for 6 consecutive days and an hour before ulcer induction.

2.3. Indomethacin induced ulcer model: Indomethacin ulcer induction was carried out according to the previously described method by Akpamu[36]. Briefly, 16 hours fasted rats received indomethacin (40 mg/kg) after 1 hour of treatments (0.5ml distilled water in A1 or folic acid in A2). Four hours later, the rats were sacrificed and laparotomy was carried out through a midline epigastric incision to isolate the stomach. The content of the stomach was obtained, and cut open (along the greater curvature) for macroscopy (using an 8MP dual camera with triple-flash light) and ulcer score. The stomach content was centrifuged and the supernatant titrated to pH 7.0 (using a 0.01 N NaOH, phenolphthalein indicator).

2.4. Pyloric ligation ulcer model: Pyloric ligation was executed following previously described Shay method[37]. Briefly, 36 hours food fasted rats were anesthetized with ketamine (0.5ml/100g) after 1 hour of treatment (0.5ml distilled water in B1 or folic acid in B2). Laparotomy was performed and the pylorus sphincter ligated with tread without damage to gastric blood supply. Four hours later, the rats were sacrificed and laparotomy was carried out through a midline epigastric incision and the stomach content drained for the determination of volume and acidity by titration as in above. Macroscopy and ulcer score were also determined as in above.

2.5. Ulcer score and indices: Gross gastric mucosal lesions was scored using the scoring method of Adinortey *et al*.[38] to determined ulcer index (UI) and the percentage protection calculated using the formula:

Protection % = [(UI control − UI treated group) / (UI control)] x 100

2.6. Sample collection and preparation: A blood sample (2ml) was obtained following cardiac puncture for the determinations of full blood count. The harvested stomachs were homogenized in 10% (w/v) of phosphate buffer (pH 7.4) and centrifuged at 4°C for 10 min. The supernatant was used for the determination of gastric tissue protein, Malondialdehyde (MDA), reduced glutathione (GSH), mucin, and nitric oxide (NO) levels.

2.7. Biochemical assays: Gastric tissue protein was determined following the Biuret method[39]. Gastric tissue MDA was determined by measuring thiobarbituric acid reactive substances (TBARS) formation according to the method of Varshney and Kale[40] and MDA level was calculated[41]. Gastric tissue GSH[42], mucin[43] and NO[44] were estimated following previously described standard methods.

2.8. Data analysis: Data analysis were done using statistical tools from SPSS (version 21) and the "student t-test" was performed at $p<0.05$ level of significance. The results were expressed as mean ± standard error of mean.

3. RESULTS

Table 1 shows the macroscopic ulcer presentation and Table 2, ulcer index and percentage protective ratio of folic acid treatment on indomethacin and pyloric ligation induced ulcers. Ulcer index was significantly reduced ($p<0.05$) in folic acid treated rats with a percentage protective ratio of 55.56% against indomethacin induced ulcer and 60.00% for pyloric ligation ulcer.

Figure 1 shows the effect of folic acid treatment on gastric acidity in indomethacin and pyloric ligation ulcers model. Folic acid treatment increased gastric acidity in both ulcer models with significant different in pyloric ligation model compared with control.

Table 3 shows the protective effect of folic acid treatment on gastric biochemical parameters in indomethacin and pyloric ligation ulcers. Folic acid treatment non-significantly protected gastric tissue protein in indomethacin induced ulceration but non-significantly destroyed gastric tissue protein in pyloric ligation ulcer. There was a non-significant decrease in gastric tissue oxidative damage in both indomethacin and pyloric ligation ulcers following folic acid treatment. Gastric tissue GSH was non-significantly increased with indomethacin induced ulcer but decreased with pyloric ligation ulcer following folic acid treatment. Folic acid treatment non-significantly increased gastric mucin level in both indomethacin and pyloric ligation ulcers. Folic acid treatment significantly increased nitric oxide level in indomethacin induced ulcer but significantly decrease it in pyloric ligation ulcer compared with control.

Table 1. Macroscopic ulcer presentation of folic acid treatment on indomethacin and pyloric ligation induced ulcers.

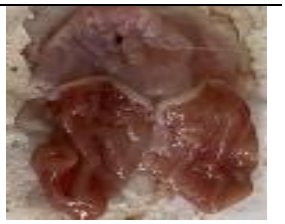
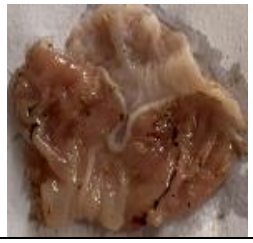
Indomethacin gastric ulcer induced		Pyloric-ligation induced	
Control	**Folic acid treated**	**Control**	**Folic acid treated**
			
			

Table 2. Ulcer index and percentage protective ratio of folic acid treatment on indomethacin and pyloric ligation induced ulcers.

	Indomethacin ulcer induced		Pyloric-ligation induced	
	Control	**Folic acid treated**	**Control**	**Folic acid treated**
Ulcer index	42.22	17.78*	37.50	15.00*
% protective ratio	0.00	55.56	0.00	60.00

Values are mean ± standard error of mean; n = 5, *significantly different compared with control.

4. DISCUSSION

The findings of the study showed that 6 days pre-treatment with folic acid presented 55.56% and 60.00% protection against indomethacin and pyloric ligation ulcers respectively. Beneficial effects of folic acid against many diseases have been extensively documented but there is a relative dearth of reports on peptic ulceration. Studies have reported pre-treatment with folic acid (2mg/kg for 21 days) prevented indomethacin induced

ulcer formation by 32%[35] and reduction in ethanol induced ulcer severity by 27% and 30% at 2 mg/kg and 3 mg/kg folic acid pre-treatment[45]. In a case report, Sharp *et al*.[46] reported reversible melanosis-like pigmentation in the duodenum associated with a gastric ulcer and folic acid deficiency.

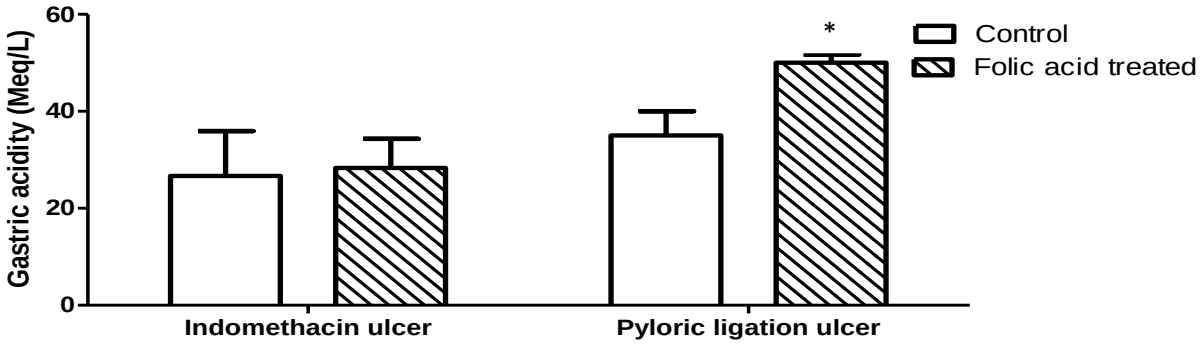


Figure 1. Effect of folic acid treatment on gastric acidity in indomethacin and pyloric ligation ulcers.

Values are mean ± standard error of mean; n = 5, *significantly different compared with control.

Table 3. Protective effect of folic acid treatment on gastric biochemical parameters in indomethacin and pyloric ligation ulcers

Gastric tissue	Indomethacin induced ulcer		Pyloric ligation ulcer	
	Control	Folic acid treated	Control	Folic acid treated
Protein (mg/dl)	1.07±0.06	1.28±0.13	1.31±0.01	1.21±0.01
MDA (u/ml)	1.16±0.001	0.59±0.003	1.19±0.01	0.89±0.001
GSH (mg/dl)	2.38±0.40	2.51±0.18	2.49±0.29	2.36±0.23
Mucin (mg/dl)	36.20±1.99	36.89±4.29	38.58±3.03	39.55±0.84
NO (ml/organ wt)	120.33±6.67	151.17±28.16*	145.89±14.86	106.86±7.64*

Values are mean ± standard error of mean; n = 5, *significantly different compared with control. Key: MDA = Malondialdehyde, GSH = Glutathione, NO = Nitric oxide

On gastric acid secretion, the present study showed that 6 days pre-treatment with folic acid stimulated gastric acid secretion compared to the control with significant different in the pyloric ligation ulcer. This finding disagrees with that of Ajeigbe *et al*.[35] who reported that 2mg/kg folic acid pre-treatment for 21 days significantly reduced basal and stimulated acid secretions in the treated when compared with control animals. In fact, the study by Ajeigbe *et al*.[35] reported that folic acid produced a 21% reduction in the basal acid output when compared with the control and 140% reduction in histamine-induced acid response. Although previous studies[35, 47] reported gastric anti-secretory properties of folic acid against indomethacin-induced gastropathy, this study's finding on acid out-put is contradictory. The difference may be due to the difference in duration of treatment with folic acid between the studies.

Recall that imbalance between the aggressive damaging and defense factors is the mechanisms for ulcer formation[10-12]. The mechanism for ulcer induction by indomethacin; an NSAID, is widely through

inhibitory activity against cyclooxygenase, which block prostaglandins production and inturn decrease mucus secretion, blood flow, disturbance of microcirculation and increase lipid peroxidation, and neutrophil activation[48] which are involved in the pathogenesis of gastric mucosal damage. The induction of gastric ulcer by indomethacin and pyloric ligation in the present study did not significantly alter mucin level compared with the folic acid pre-treated groups. This findings indicate that folic acid may not significantly affect the mucus protective mechcanism of the stomach. However, a study have shown folic acid treatment to elevate mucin concentration by 24% following indomethacin ulcer induction[49]. MDA levels were decreased in folic acid pre-treated groups as compared to the indomethacin and pyloric ligation ulcer group but to a non-significant level. The non-significant degree of mucin, MDA and GSH levels observed in the present study may be due to the period of pre-treatment. Variations exist in the fate of folic acid mechanism in gastric ulcer protection in the two models.

In the indomethacin model, folic acid induced protection of the gastric mucosa was via inhibition of oxidative lipid peroxidation, stimulated gastric antioxidant and nitric oxide levels while in the pyloric ligation model is via inhibition of oxidative lipid peroxidation and increase mucin production. In accordance with the findings of this study, Ajeigbe *et al*.[35] has also documented indomethacin-induced lipid peroxidation was attenuated by 58% reduction in MDA concentration on pretreatment with folic acid pre-treatment. The observed protective effect of folic acid against gastric ulcers may be due to the noted reported that folate modulates a number of disorders as a result of its anti-apoptotic and anti-oxidative properties[26].

5. CONCLUSION

Conclusively, folic acid pre-treatment protects the gastric mucosa against indomethacin and pyloric ligation ulcers by inhibiting oxidative damage through increased production of antioxidant (GSH), mucus and elevated blood flow via stimulating nitric oxide level in the indomethacin ulcer model. However, in the pyloric ligation ulcer model, protection of the gastric mucosa is through decreased oxidative stress and increased mucus secretion.

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